January 2026 CSE 316: Online on Assembly Language
Subsection: A2
Time: 35 minutes + Submission
July 6, 2026

Problem: Bubble Sort an Array

There is an array which contains positive integers. Your task is to sort the array in ascending order using the Bubble Sort algorithm and print the sorted array. You have to:

1. Take the size of the array as input.
2. Take the array elements as input. The number of elements must be equal to the size provided.
3. Sort the array in ascending order using Bubble Sort.
4. Print the sorted array.

You can assume that the size of the array and each array element consist of a single digit. That is, for the array size n, $0 < n < 10$, and for each array element e, $0 \leq e < 10$. You must implement the sorting logic using loops and conditional jumps in 8086 assembly language.

Bubble Sort Algorithm in C

The following C code shows the logic that you need to implement in 8086 assembly language.

```
int arr[10];
int n;

// take n as input
// take n array elements as input

for (int i = 0; i < n - 1; i++) {
    for (int j = 0; j < n - i - 1; j++) {
        if (arr[j] > arr[j + 1]) {
            int temp = arr[j];
            arr[j] = arr[j + 1];
            arr[j + 1] = temp;
        }
    }
}
```

Examples

Example 1
Input:

```
8
53814726
```

Output:

```
12345678
```

Example 2
Input:

```
9
925173840
```

Output:

```
012345789
```